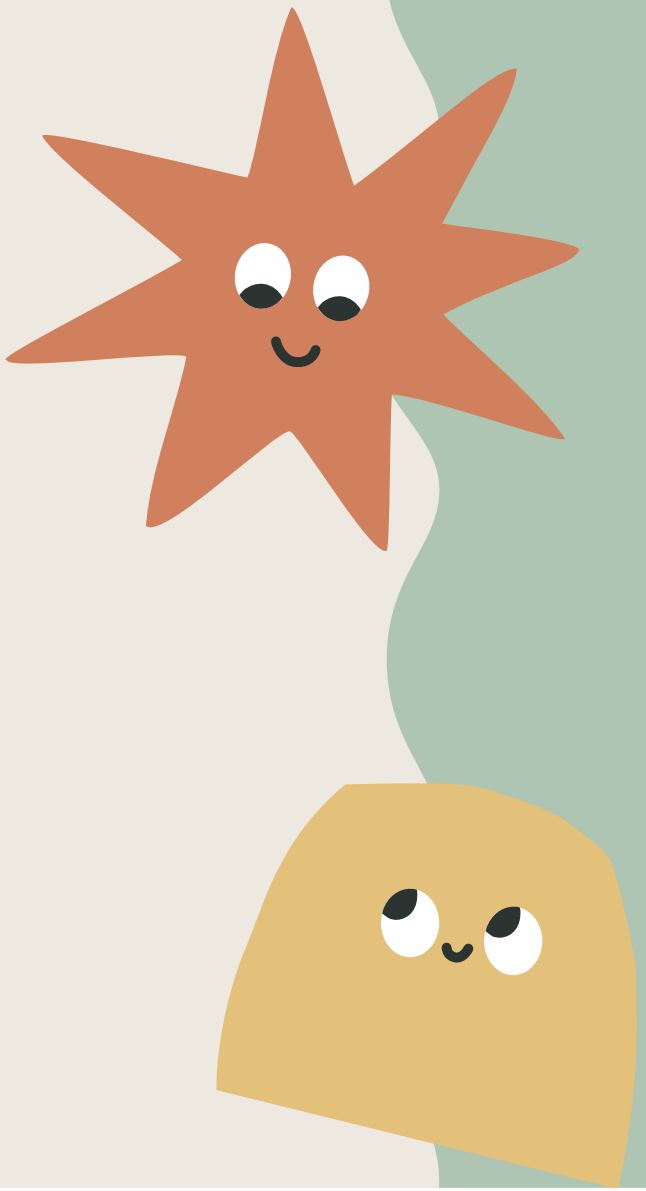
Reasoning About Action Feasibility from Physical Properties



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Sep 21, 2026

幻灯片 1

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Initial Average Score: 4.6/5
-0.5 for late submission
Final Score: 4.1/5
Good length, but if you have to say that the text on your diagrams is too small you might want to break it across multiple slides so you can zoom in
Danielle Villa, 2026/9/22

Oops, No Knife in the Kitchen Again!

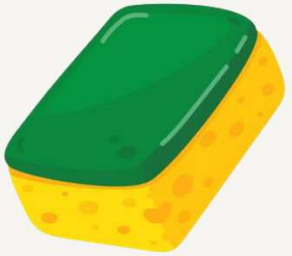
Goal: Cut a carrot



1. Plastic Card



2. Rolling Pin



3. Sponge

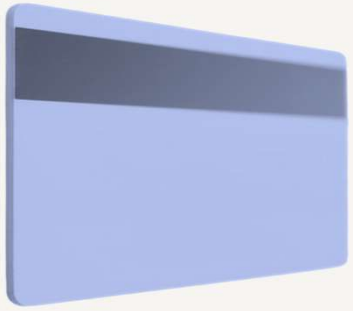
- Imagine that a conventional cutting tool is unavailable:

You have only three objects: a plastic card, a rolling pin, and a sponge.

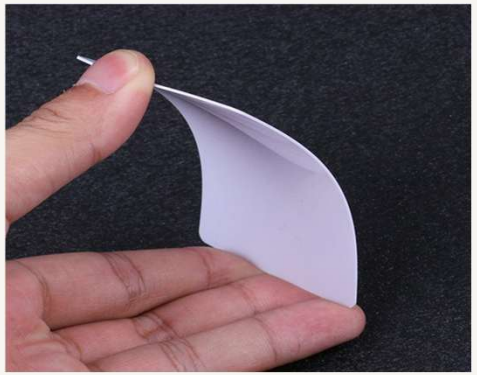
- **Which object could physically support cutting?**
  And how can we explain why it would work?

Same Object Name, Different Physical Properties

- You might think, “Why not just ask an LLM?” But simply answering “plastic card” is not enough.



A rigid card
Its edge may satisfy the physical requirements for Cut.



A flexible card
The same object category may no longer satisfy those requirements.

- Even within the same object category, physical properties can differ. A rigid card and a flexible card may have very different cutting capabilities.
- **So, instead of relying on object names, we need to reason about the physical properties that make cutting possible.**

Our Course Project: A Focused Scope

One action: CUT
Reasoning about cutting feasibility from physical properties

In Scope
- One action: Cut
- Evaluation example: Cutting a carrot
- Ordinary human actor in a kitchen
- Already represented object-part properties
- Property–requirement reasoning

Out of Scope
- Other actions (Open, Pour, Squeeze)
- Identifying properties from raw inputs
- General action–target applicability
- Robots, vision, and object recognition
- Broader environments and industrial tools

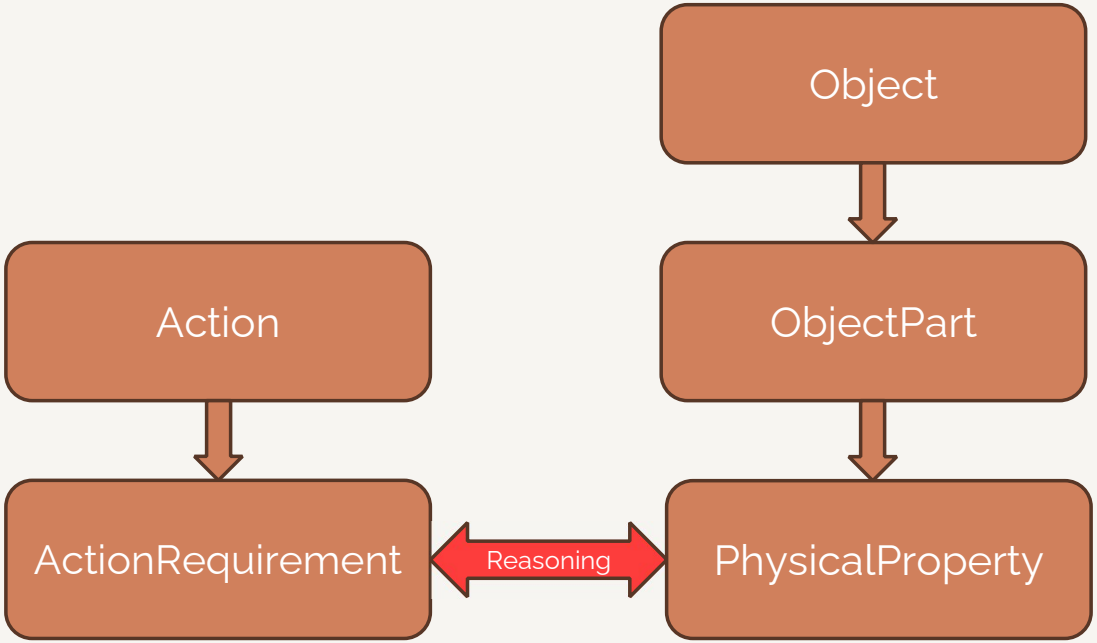
Our main goal: Define the physical requirements for Cut and formalize their relationships with object-part properties.

What our ontology does: Given already represented physical properties, evaluate whether they satisfy the cutting requirements and explain the result.

What it does not do: Identify properties from raw images or determine action applicability to arbitrary targets. These belong to a larger system.

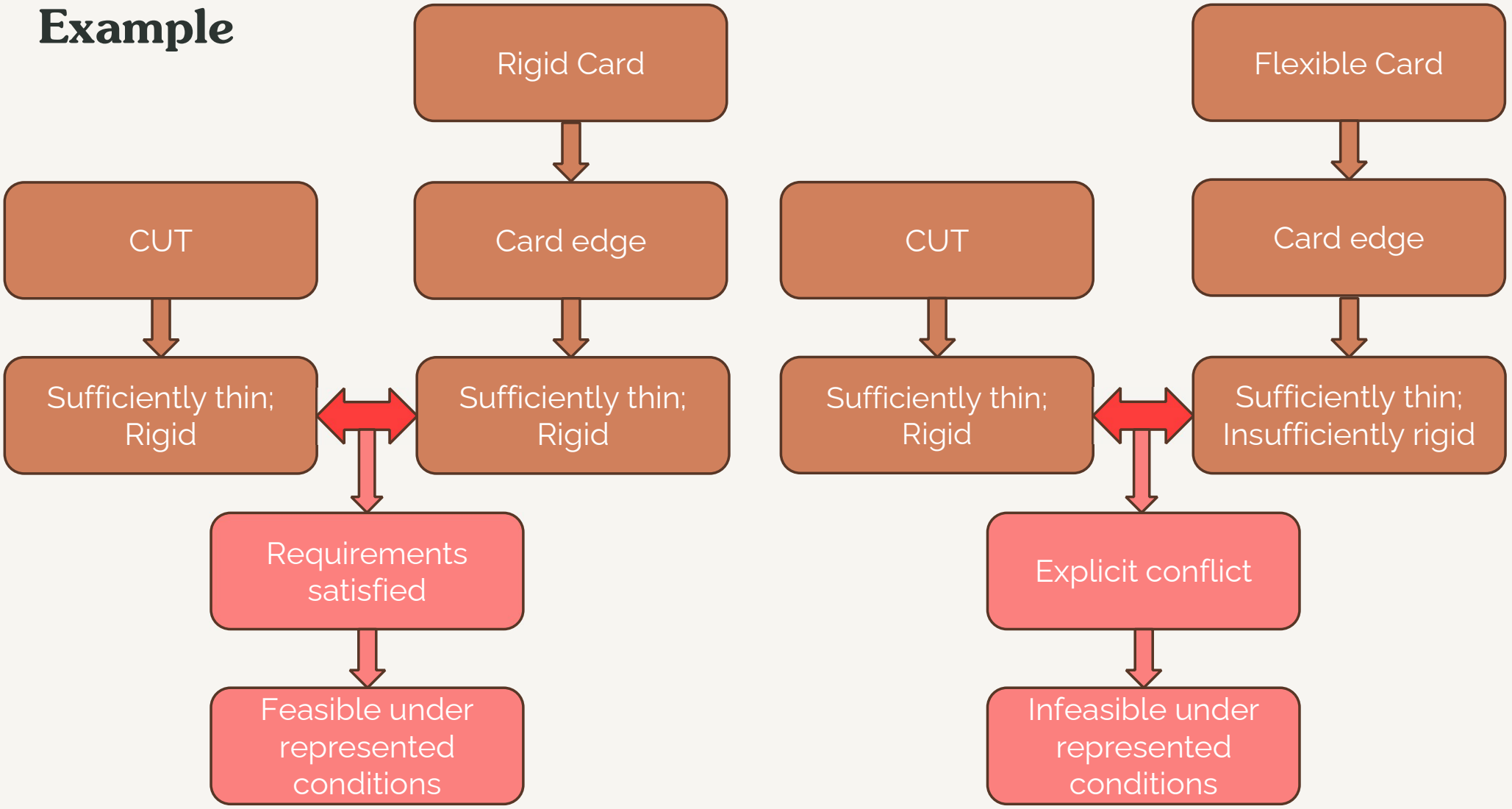
* We will update our written use case accordingly.

Core Ontology Idea



- The ontology represents two connected structures, as shown on the left.
- Reasoning:
  - What object can support Cut?
  - Is the object incompatible for the action or information that the current ontology has is insufficient?
  - Provide a property-based explanation for the result.

Example


Rigid Card
CUT
Card edge
Sufficiently thin; Rigid
Sufficiently thin; Rigid
Requirements satisfied
Feasible under represented conditions
Flexible Card
CUT
Card edge
Sufficiently thin; Rigid
Sufficiently thin; Insufficiently rigid
Explicit conflict
Infeasible under represented conditions

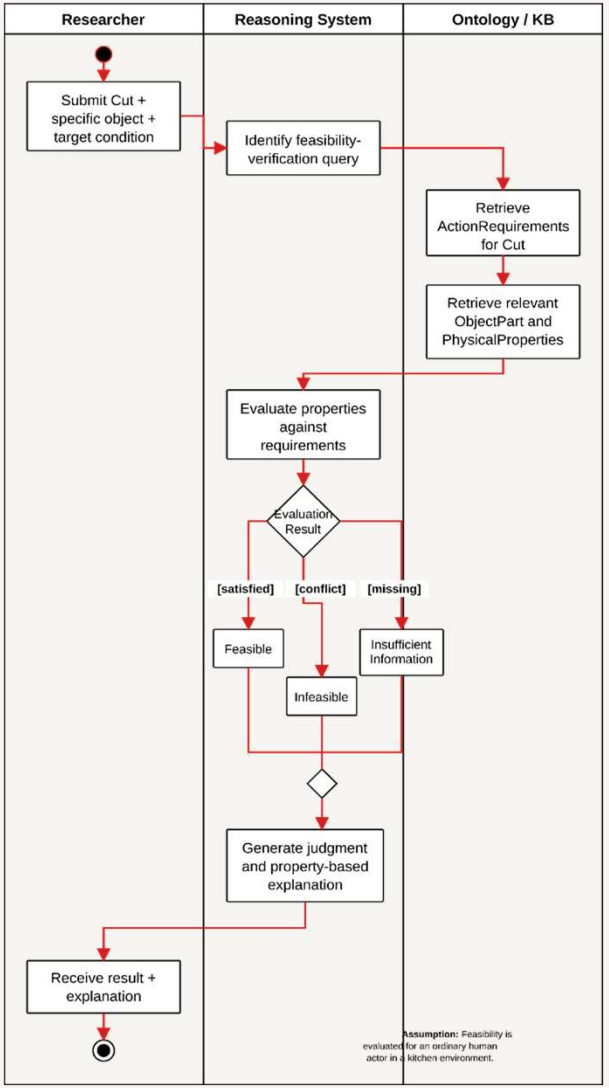
Actors and Interaction

Researcher	Ordinary Human Actor	Ontology-Based Reasoning System	Knowledge Base/ Data Source
Submits queries to the ontology-based reasoning system. Receives the resulting feasibility judgment, candidate objects, and property-based explanations.	Represents the assumed physical capabilities of the person who would perform the action.	Uses the ontology to retrieve the requirements associated with Cut, identify relevant object parts and their represented physical properties, and reason about whether those properties satisfy or conflict with the action requirements.	Supplies structured information used to populate or support the ontology, including objects, object parts, physical properties, and evidence used to define or refine cutting requirements.

Usage Scenario 1:

Feasibility Verification

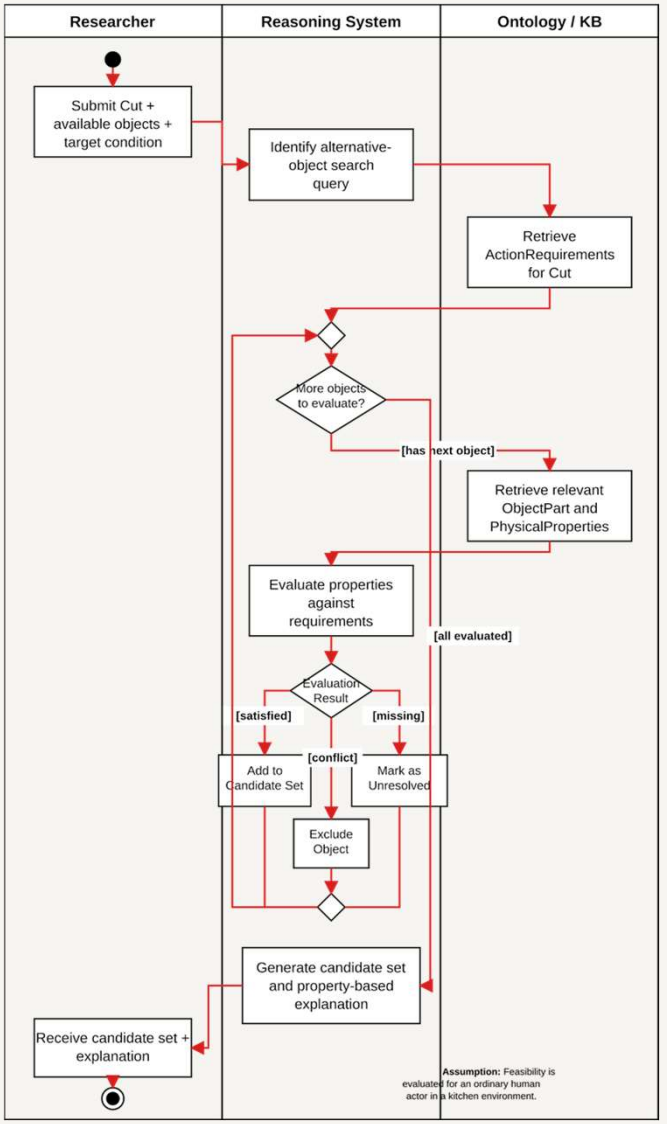
- A researcher studying physical affordances wants to determine whether a specific everyday object could be used by an ordinary human actor to perform **CUT** in a kitchen setting.
    - ″Can a rigid plastic card could be used to cut a carrot″
- The system uses the ontology to:
    - **Retrieve** the physical requirements represented for cutting
    - **Identify** the relevant part of the card
    - **Retrieve** the physical properties associated with that part.
- It then determines whether those represented properties **SATISFY** or **CONFLICT** with the cutting requirements.
- The desired outcome is a **feasible**, **infeasible**, or **insufficient-information** judgment
    - Paired with an **explanation** identifying the relevant object part, physical properties, and physical requirements that support the result.



Usage Scenario 2:

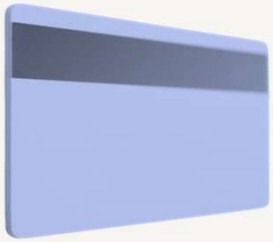
Goal-Driven Alternative Object Discovery

- A researcher studying alternative object use wants to identify which objects from a known set of available kitchen items could support **CUT** when a conventional cutting tool is unavailable or unusable.
    - For example: the available objects may include a rigid plastic card, a rolling pin, and a kitchen sponge.
- Rather than searching for objects conventionally categorized as cutting tools:
    - the system starts from the **physical requirements** represented for cutting
    - The system uses the ontology to examine the **relevant parts and properties** of each available object.
- Objects whose represented properties satisfy the cutting requirements are **RETURNED as candidates**
    - Objects with explicitly incompatible properties are **EXCLUDED**.
- The desired outcome is a set of **physically plausible alternative objects**
    - paired with property-based explanations for why each candidate satisfies the represented cutting requirements.



Can Our Ontology Answer a Concrete Question?

- **CQ1.1:** Given the represented properties of a plastic card, can an ordinary human actor use it to cut a carrot, and why?



Represented object-part properties

Plastic Card
↓
Card Edge
↓
(Thin + sufficient rigidity)



Cut requirements (target: carrot)

Cut
↓
ActionRequirements
↓
(Thin + sufficient rigidity)

Ontology approach

1. Given the represented properties of the card,
2. Retrieve the ActionRequirements for Cut.
3. Compare the properties with the requirements.

Example answer: Feasible under the represented conditions, because the card edge satisfies the relevant cutting requirements.

Can We Actually Build This?

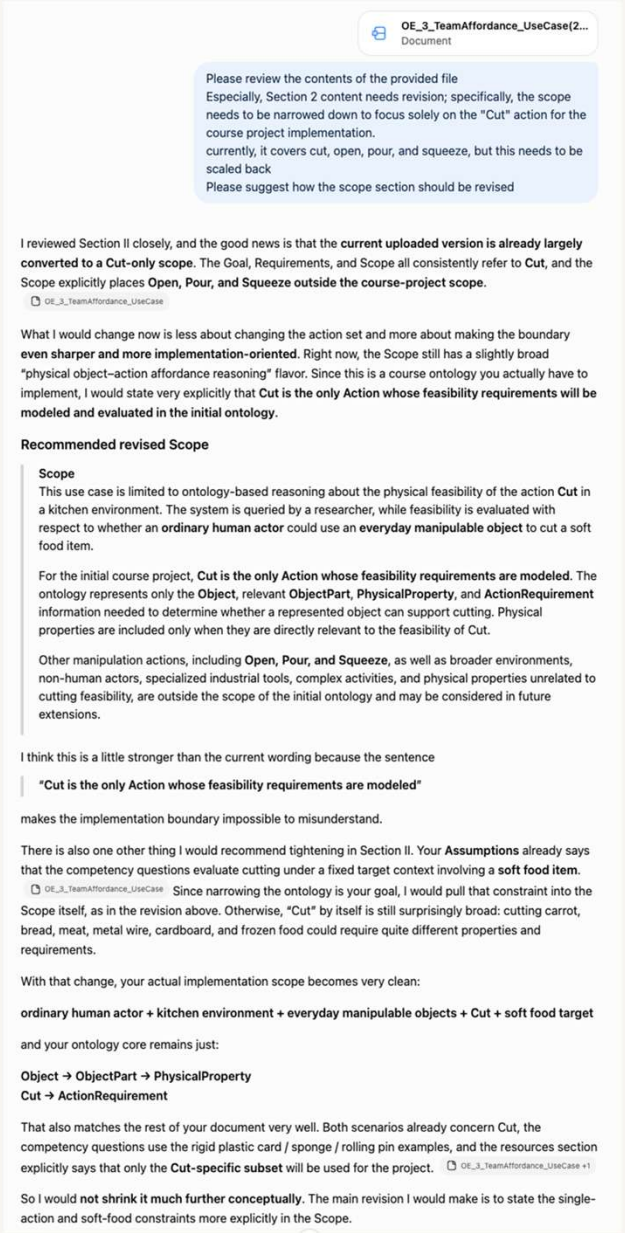
- A focused ontology supported by curated data

Resource	Role in our project
EPIC-KITCHENS-100	Kitchen action vocabulary and source context for Cut
Feasibility annotations	Object parts, physical properties, feasible/infeasible examples
Counterfactual annotations	Supporting examples for refining physical requirements
WordNet + Relations Ontology	Terminology normalization and potential reuse of existing relations

- **Data:** A curated Cut-specific subset will supply the object-part properties and feasibility examples.
- **Implementation:** Represent the necessary ontology relations and test our four competency questions.
- **Main challenge:** Formalizing property–requirement compatibility while distinguishing explicit conflict from missing information.

GenAI Use + What Changed

- Model: ChatGPT 5.6
- **GenAI was used to:**
  - Refine the scope to Cut only.
  - Clarify Researcher vs. Ordinary Human Actor.
  - Plan the UML activity diagrams.
- **Further clarification from team discussion:**
  - Represented physical properties are assumed as input.
  - Our ontology evaluates property–requirement compatibility rather than identifying properties from raw objects.



OE_3_TeamAffordance_UseCase(2...
Document

Please review the contents of the provided file
Especially, Section 2 content needs revision; specifically, the scope needs to be narrowed down to focus solely on the "Cut" action for the course project implementation.
currently, it covers cut, open, pour, and squeeze, but this needs to be scaled back
Please suggest how the scope section should be revised

I reviewed Section II closely, and the good news is that the **current uploaded version is already largely converted to a Cut-only scope**. The Goal, Requirements, and Scope all consistently refer to **Cut**, and the Scope explicitly places **Open, Pour, and Squeeze outside the course-project scope**.
OE_3_TeamAffordance_UseCase

What I would change now is less about changing the action set and more about making the boundary **even sharper and more implementation-oriented**. Right now, the Scope still has a slightly broad "physical object–action affordance reasoning" flavor. Since this is a course ontology you actually have to implement, I would state very explicitly that **Cut is the only Action whose feasibility requirements will be modeled and evaluated in the initial ontology**.

Recommended revised Scope

> **Scope**
> This use case is limited to ontology-based reasoning about the physical feasibility of the action **Cut** in a kitchen environment. The system is queried by a researcher, while feasibility is evaluated with respect to whether an **ordinary human actor** could use an **everyday manipulable object** to cut a soft food item.
>
> For the initial course project, **Cut is the only Action whose feasibility requirements are modeled**. The ontology represents only the **Object**, relevant **ObjectPart**, **PhysicalProperty**, and **ActionRequirement** information needed to determine whether a represented object can support cutting. Physical properties are included only when they are directly relevant to the feasibility of Cut.
>
> Other manipulation actions, including **Open, Pour, and Squeeze**, as well as broader environments, non-human actors, specialized industrial tools, complex activities, and physical properties unrelated to cutting feasibility, are outside the scope of the initial ontology and may be considered in future extensions.

I think this is a little stronger than the current wording because the sentence

> **"Cut is the only Action whose feasibility requirements are modeled"**

makes the implementation boundary impossible to misunderstand.

There is also one other thing I would recommend tightening in Section II. Your **Assumptions** already says that the competency questions evaluate cutting under a fixed target context involving a **soft food item**.
OE_3_TeamAffordance_UseCase Since narrowing the ontology is your goal, I would pull that constraint into the Scope itself, as in the revision above. Otherwise, "Cut" by itself is still surprisingly broad: cutting carrot, bread, meat, metal wire, cardboard, and frozen food could require quite different properties and requirements.

With that change, your actual implementation scope becomes very clean:

ordinary human actor + kitchen environment + everyday manipulable objects + Cut + soft food target

and your ontology core remains just:

Object → ObjectPart → PhysicalProperty
Cut → ActionRequirement

That also matches the rest of your document very well. Both scenarios already concern Cut, the competency questions use the rigid plastic card / sponge / rolling pin examples, and the resources section explicitly says that only the **Cut-specific subset** will be used for the project. OE_3_TeamAffordance_UseCase +1

So I would **not shrink it much further conceptually**. The main revision I would make is to state the single-action and soft-food constraints more explicitly in the Scope.